

## Formula

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}$$

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \quad \$\$$$

## Example

Scores example: difference = 4, SE =  $\sqrt{5^2/40 + 6^2/35} \approx$

$$1.32, \quad 95\% \text{ CI} = 4 \pm 1.960 \cdot 1.32 \Rightarrow (1.4, 6.6).$$

## Interpretation

If 0 is outside the CI, the two-tailed test at  $\alpha = 0.05$

rejects  $H_0$ .